that lethal force could be employed without meaningful human oversight, contradicting ethical norms and risking unintended harm due to algorithmic misinterpretation or data errors.

A central issue is the “accountability gap” determining who is responsible when an autonomous system makes a faulty or harmful decision. Is it the programer, the commanding officer, the political authority, or the machine’s internal logic? Current legal systems are not fully equipped to address these distributed layers of responsibility. Additionally, the proliferation of low-cost autonomous weapons poses geopolitical risks, as non-state actors or unstable regimes may acquire systems capable of targeted violence without adequate oversight or regulation.

To mitigate these concerns, international organizations and research institutions emphasize the need to maintain meaningful human control (MHC) over all lethal operations. This requires that humans remain responsible for final engagement decisions, and that autonomous systems function as advisory or assistance tools rather than independent agents. There is also growing momentum toward international frameworks governing AI transparency, algorithmic accountability, kill-switch protocols, and autonomous weapons treaties. Establishing these norms will be essential to ensure that advancements in drone warfare align with humanitarian principles and global security stability.

Table 8.3 outlines the principal ethical and legal challenges associated with the increasing autonomy of drone warfare systems. As UAVs gain capabilities to select and engage targets with limited human oversight, fundamental concerns arise regarding *accountability, transparency, and adherence to international humanitarian norms*. The automation of lethal decision-making challenges longstanding ethical expectations that human